

Data Monitoring tool

- Features in current project:

1. **Dataset Upload**

- a. CSV
- b. Excel
- c. JSON
- d. Parquet
- e. TSV
- f. TXT
- g. Pickle

2. **Dataset Overview**

- a. Number of rows and columns
- b. Column names
- c. Data types
- d. Basic dataset information

3. **Data Preview**

- a. View uploaded records
- b. Inspect the dataset before analysis

4. **Data Quality Analysis**

- a. Missing-value detection
- b. Duplicate detection
- c. Basic consistency analysis

5. **Statistical Analysis**

- a. Mean
- b. Median
- c. Standard deviation
- d. Minimum/maximum
- e. Descriptive statistics

6. **Correlation Analysis**

- a. Correlation matrix
- b. Correlation visualization

7. **Outlier Detection**

- a. IQR-based outlier detection

- b. Identification of unusual values

8. **Anomaly Detection**

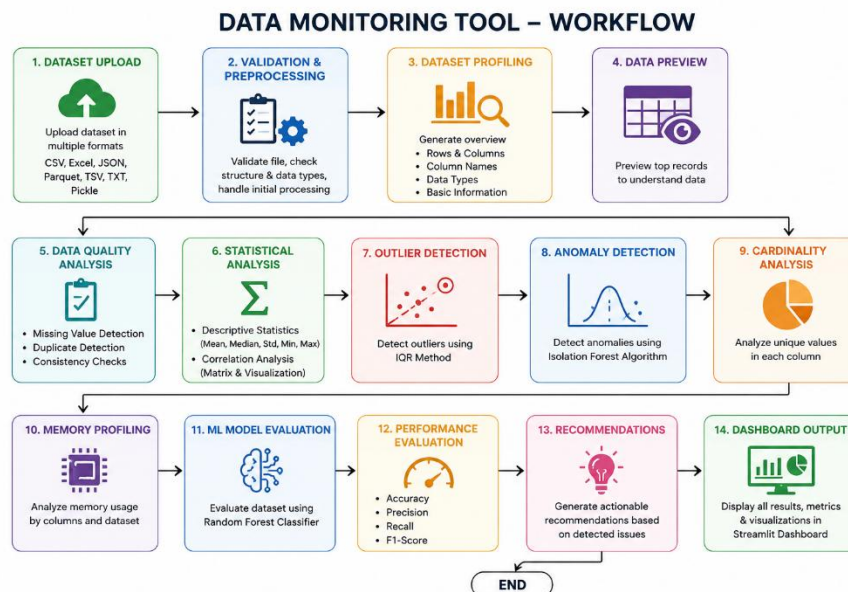
- a. **Isolation Forest**
- b. Detection of anomalous records

9. **Cardinality Analysis**

- a. Unique-value counts
- b. Identification of high-cardinality columns

10. **Memory Profiling**
 - a. Memory usage analysis
 - b. Identification of memory-heavy columns
11. **ML Model Evaluation**
 - a. Random Forest classifier
 - b. Accuracy
 - c. Precision
 - d. Recall
 - e. F1-score
12. **Visual Analytics**
 - a. Statistical visualizations
 - b. Correlation visualization
 - c. Outlier/anomaly visualizations
13. **Automated Recommendations**
 - a. Recommendations based on detected data-quality issues
14. **Database Connectivity**
 - a. SQLAlchemy
 - b. PostgreSQL integration

Workflow:



Changes to be done:

Phase	Features to Implement	Goal
1	Schema monitoring, quality rules, freshness, volume, data drift	Strong monitoring engine

2	Continuous monitoring, history, health score, alerts	Data observability
3	Lineage, root-cause analysis, impact analysis	Enterprise capabilities
4	AI diagnosis, AI recommendations, remediation	AI-powered monitoring
5	Authentication, API, testing, Docker, CI/CD, deployment	Production readiness

More advance features:

Technology	How you could use it	Difficulty	Recommendation
Generative AI / LLM	Explain anomalies, generate SQL/data-quality rules, summarize incidents, suggest fixes	Medium	Definitely add
Explainable AI (XAI)	Explain <i>why</i> an anomaly/model prediction was flagged using SHAP/LIME	Medium	Definitely add
Digital Twin	Create a virtual representation of your dataset/data pipeline and simulate quality/drift problems	High	Excellent advanced feature
📶 Edge AI	Monitor data directly on IoT/edge devices before sending it to the central platform	High	Add only if you use IoT/streaming data
AI Agents / Agentic AI	Agent investigates an incident → finds cause → recommends/executes remediation	High	Very strong 2026 feature
🔗 Knowledge Graph	Connect datasets, columns, pipelines, models and incidents for intelligent root-cause analysis	High	Advanced option
🔊 Real-time Streaming	Monitor Kafka/MQTT/event streams continuously	High	Excellent for industry use

Final Workflow:

